

LOKESH R M

AI / Robotics Engineer · Physical AI, Agentic AI, Computer Vision, ROS 2

+91 9035446966 | hi.lokeshrm@gmail.com | lokeshrm.work@gmail.com | Bengaluru, Karnataka, India
linkedin.com/in/thisislokeshrm | github.com/hilokeshrm | hilokeshrm.github.io

SUMMARY

AI and robotics engineer at Sirena Technologies working on both sides of physical AI: humanoid robot autonomy and the agent systems that direct it. Architected a 53 service multi agent platform running at 3 enterprise clients, and build the vision, speech and ROS 2 navigation stack for biped and quadruped robots. Lead teams of 3 to 5 across agent infrastructure, robot cloud services and perception.

EXPERIENCE

AI Engineer - Sirena Technologies, Bengaluru, India

Jul 2026 - Present

- Architected **Agent OS**, a 53 service multi agent platform deployed to 3 enterprise clients, and designed and own its **Policy Gateway** enforcing 153 policies for centralised agent authorisation and monitoring. Platform team of 12, gateway team of 3.
- Led development of 5 in house agents across opportunity tracking, ROI, marketing and finance workflows, and 3 third party agent integrations.
- Led a 5 person team building **Sirena Cloud**, the voice, vision and LLM service layer supporting 80+ concurrent robots on an NVIDIA RTX Pro 6000.
- Built the streaming voice pipeline behind human robot interaction, combining VAD, ASR, TTS and LLM reasoning at roughly 60 ms response latency.
- Fine tuned YOLO and SSD detectors for office and industrial environments, and optimise inference with ONNX for NVIDIA Jetson Orin Nano.
- Deploy ROS 2 Nav2 and SLAM navigation for autonomous service robots and a 4 foot autonomous biped, and self host open weight LLMs with RAG retrieval on pgvector.

Robotics Engineer, Graduate Trainee - Sirena Technologies, Bengaluru and Solan, India

May 2025 - Jun 2026

- Built MATLAB and Python locomotion simulations for biped and quadruped robots covering walk sequences and control algorithms, and validated every platform in simulation before hardware build.
- Developed features in the LS6 customised robot controller codebase and implemented ROS 2 motion control for robotic arms and wheeled robots.
- Led a 3 person team building the cloud system for an ESP32 P4 based home robotics product, including ESP-IDF firmware integration.
- Completed the graduate engineering rotation at the Solan facility and converted to a full time engineering role.

Machine Learning Intern - MURTI Labs, GITAM Bengaluru, India

May 2024 - Apr 2025

- Trained LSTM based object detection and action recognition models on drone video to 83% accuracy, and implemented path mapping for autonomous delivery and emergency response drones.

Freelance Full Stack and AI Engineer - Independent, Bengaluru, India

2024 - Present

- Delivered production web and AI applications for 5+ clients, covering LLM integration, agent workflows, APIs and backend infrastructure.

SKILLS

AI and Agents: LLMs, RAG, Multi Agent Systems, Agent Orchestration, LangGraph, LangChain, CrewAI, AutoGen, MCP, pgvector, Open Weight Model Hosting

Vision and Speech: YOLO, SSD, OpenCV, MediaPipe, ONNX, ASR, TTS, Voice Activity Detection, PyTorch, TensorFlow, LSTM

Robotics: ROS 2, Nav2, SLAM, Navigation, Motion Control, Biped and Quadruped Locomotion, Control Algorithms, State Estimation, Sensor Integration, Sim2Real

Simulation: MATLAB, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, Webots

Edge and Infrastructure: NVIDIA Jetson Orin Nano, RTX Pro 6000, ESP32 P4, ESP-IDF, Docker, Kubernetes, FastAPI, Kafka, Redis, Google Cloud, PostgreSQL, Linux, Git

Programming: Python, C++, C, MATLAB, SQL, JavaScript

EDUCATION

B.Tech, Computer Science and Engineering - GITAM (Deemed to be University), Bengaluru

Aug 2022 - Aug 2026

CGPA: 7.9 / 10

ACHIEVEMENTS

- INSPIRE Award MANAK** (Government of India, state level) · **Young Scientist Award** (Government of Karnataka, district level) · **Rajya Puraskar** (Governor's Award) · Finalist, Internal Smart India Hackathon, GITAM
- Secretary, IEEE Computer Society GITAM** (Dec 2024 - Aug 2025) · Student Placement Coordinator, GITAM Career Guidance Cell